

Model Portfolio Summary: John Doe

The model portfolio optimizer was required to have a max of 70% stocks, a minimum of 20% bonds, and a 20% max holding. Also to stay inside his 20% drawdown limit. It succeeded in that the worst fall was 19.1%. But it only earned 7.0% a year against 10.3% for a 60/40 benchmark.

Results

Over the backtest (Oct 4, 2019 to Sep 25, 2026), the model earned less than the benchmark but with less bumpiness and shallower falls, calming volatility.

Measure	Model portfolio	60/40 benchmark
Annual return	7.0%	10.3%
Volatility	8.3%	12.7%
Sharpe ratio	0.62	0.68
Worst drawdown	-19.1%	-22.1%
COVID crash return (Feb 19 - Mar 23, 2020)	-9.9%	-21.8%
2022 return (full year)	-14.4%	-16.3%

Did it meet John's goals

It passed two of three tests but missed the return target.

- **(Pass?) Drawdown limit:** The worst fall was 19.1% against a 20% limit. The plain 60/40 benchmark broke the limit at 22.1%.
- **(Fail) Beat the benchmark's Sharpe ratio:** 0.62 against 0.68. The model was smoother but did not earn enough extra return per unit of risk to make up for it.
- **(Pass) Stress periods:** It lost about half as much as the benchmark in the 2020 crash and less in 2022, when stocks and bonds fell together.
- **(Fail?) Return target:** The 6-7% goal was after inflation, and a 7.0% nominal return is probably below that.

What went wrong

The optimizer trusted the recent past too much, and the results leave out real-world costs.

- **It kept maxing out its favorites.** VOO sat at the 20% cap in six of the seven yearly rebalances.
- **The drawdown margin was thin.** The optimizer aimed for 15% on its data, yet the real worst fall reached 19.1%.
- **The benchmark won on return.** The period was strong for US stocks, so holding less stock cost return in exchange for a smoother ride.
- **Simplifications.** No taxes, no trading costs, one fixed 2% risk-free rate, and only about seven years of walk-forward results. So there really wasn't enough data.

Method

Each year the optimizer used only the previous three years of data to choose weights, then held them for one year, so it never saw the period it was tested on. Forcing it not to cheat.

- **Assets:** 11 ETFs across US and international stocks, bonds, real estate, gold, and short-term Treasuries.
- **Objective:** highest Sharpe ratio, the return earned per unit of risk.
- **Rules:** long only, no holding above 20%, stocks at most 70%, bonds at least 20%, and a strict drawdown limit applied on the data.
- **Benchmark:** 60% VOO and 40% BND, rebalanced once a year on the same dates.